03 · RESULTS & DISCUSSION

Findings & Analysis

A synthesis of usability testing results from **40 respondents**, heuristic evaluation findings, and data-driven recommendations for improving the **Tigaon – Primary Hospital and Clinic Website**.

40	73.1	4.33	2.00
Respondents	SUS Score	Best Item (Q5)	Lowest Item (Q6)

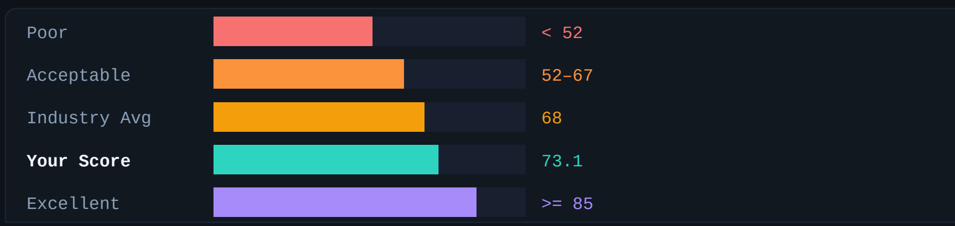
SYSTEM USABILITY SCALE RESULTS

Overall SUS Score

Scores were calculated using the standard SUS formula: odd items → (rating – 1); even items → (5 – rating). The sum of all 10 contributions per respondent is multiplied by 2.5, yielding a final score of 0–100.

73.1

Grade B · Good Usability



Above Industry Average

Exceeds the 68 industry benchmark. Perceived as usable with targeted areas for improvement.

RAW SURVEY DATA

SUS Response Frequency Table

Number of respondents (out of 40) selecting each rating per SUS statement. Data sourced directly from *SurveyResult.xlsx*. "Contrib." = average per-respondent SUS contribution per item.

No.	Statement	SD 1	D 2	N 3	A 4	SA 5	Total	Mean	Contrib.
01	I think that I would like to use this system frequently.	1	—	5	24	10	40	4.05	3.050
02	I found the system unnecessarily complex.	2	18	16	2	2	40	2.60	2.400
03	I thought the system was easy to use.	—	—	2	29	9	40	4.17	3.175
04	I think that I would need the support of a technical person.	3	25	10	2	—	40	2.27	2.725
05	I found the various functions in this system were well integra…	—	—	1	25	14	40	4.33	3.325
06	I thought there was too much inconsistency in this system.	6	28	6	—	—	40	2.00	3.000
07	I would imagine that most people would learn to use this syste…	—	—	5	24	11	40	4.15	3.150
08	I found the system very cumbersome to use.	7	27	4	2	—	40	2.02	2.975
09	I felt very confident using the system.	—	—	6	26	8	40	4.05	3.050
10	I needed to learn a lot of things before I could get going.	3	20	10	4	3	40	2.60	2.400
	SUM × 2.5 = SUS Score								73.1

Note: SD = Strongly Disagree, D = Disagree, N = Neutral, A = Agree, SA = Strongly Agree. "—" = zero responses. Positive items (Q1,3,5,7,9): Contrib = Mean – 1. Negative items (Q2,4,6,8,10): Contrib = 5 – Mean.

HEURISTIC EVALUATION

Nielsen's 10 Heuristics — Violation Summary

All 5 group members independently evaluated the prototype using Jakob Nielsen's 10 Usability Heuristics. The table below aggregates violations found across all evaluators.

#	Heuristic	Violations	Severity
H1	Visibility of System Status	2	Medium
H2	Match Between System and Real World	1	Low
H3	User Control and Freedom	3	High
H4	Consistency and Standards	2	Medium
H5	Error Prevention	2	Medium
H6	Recognition Rather Than Recall	1	Low
H7	Flexibility and Efficiency of Use	2	Medium
H8	Aesthetic and Minimalist Design	3	High
H9	Help Users Recognize & Recover from Errors	1	Low
H10	Help and Documentation	2	Medium

KEY ISSUES IDENTIFIED

Usability Problems

Issues identified through heuristic evaluation and open-ended feedback from 40 respondents, cross-referenced with low-scoring SUS items. Sorted by severity.

HIGH

No Feedback on Form Submission

After submitting forms, users received no visible confirmation, leaving them uncertain whether submissions were received. Open-ended responses corroborated this. Aligns with H1 violations identified by all 5 evaluators.

Violates H1: Visibility of System Status · H9: Help Users Recover from Errors

HIGH

Cluttered Homepage Layout

The homepage presents too many competing elements, burying the primary call-to-action. Despite Q5 scoring highest (avg 4.33 — functions well integrated), discoverability remains a barrier — consistent with H8 violations flagged by all evaluators.

Violates H8: Aesthetic and Minimalist Design · H3: User Control and Freedom

HIGH

No Bikol / Filipino Language Support

Multiple respondents explicitly recommended Bikol or Filipino/Tagalog translation, noting elderly community members and non-English speakers would struggle with the platform. This is a critical accessibility gap for a community health service.

Violates H2: Match Between System and Real World · H7: Flexibility and Efficiency of Use

MEDIUM

Insufficient Doctor/Specialist Directory

Respondents noted the specialist directory lacked availability information and role details, making it feel unhelpful. Reflects an H2 violation where the system does not match real-world hospital directory expectations.

Violates H2: Match Between System and Real World · H6: Recognition Rather Than Recall

MEDIUM

No Back Navigation on Multi-Step Flows

During appointment scheduling, users cannot edit previous steps without restarting. Q10 (avg 2.60) had 7 respondents selecting Neutral or Agree — indicating a learnability barrier for users without prior experience with similar systems.

Violates H3: User Control and Freedom · H5: Error Prevention

LOW

Missing Onboarding and Help Documentation

Respondents suggested a step-by-step tutorial for new users. Aligns with H10 violations and Q4 (avg 2.28) — 12 of 40 respondents were neutral or agreed they needed technical support to use the system.

Violates H10: Help and Documentation · H6: Recognition Rather Than Recall

DISCUSSION OF FINDINGS

What the Data Tells Us

The average SUS score of **73.1** places the Tigaon – Primary Hospital and Clinic Website in the **"Good" usability range (Grade B)**, exceeding the industry benchmark of 68. This indicates the system is generally usable for its intended audience — 40 respondents predominantly composed of 2nd-year education students aged 19–26 from Partido State University.

The highest-performing items were **Q5** ("Functions well integrated," avg 4.33), **Q3** ("Easy to use," avg 4.18), and **Q7** ("Most people would learn quickly," avg 4.15). Negatively worded items also performed well: **Q6** ("Too much inconsistency," avg 2.00) and **Q8** ("Cumbersome to use," avg 2.03) had very low means — meaning 28 and 27 of 40 respondents respectively *disagreed* with these negative statements.

Despite the positive SUS outcome, **Q10** ("Needed to learn a lot," avg 2.60) and **Q2** ("Unnecessarily complex," avg 2.60) showed the widest spread — signaling a learnability gap for a subset of users, particularly the 20 of 40 respondents who had no prior experience with similar systems.

Qualitative open-ended responses significantly enriched the quantitative findings. The most consistent theme was **the need for local language support** (Bikol/Filipino) — raised independently by multiple respondents and not captured by any SUS item. The second most common theme was the need for more detailed specialist information and better booking confirmation feedback. Addressing these specific gaps would realistically push the SUS score into the Excellent range (>= 85).

RECOMMENDATIONS

Suggested Design Improvements

Based on SUS data from 40 respondents, open-ended qualitative feedback, and heuristic evaluation findings — ranked by priority.

01 **Add Bikol and Filipino Language Support**
Implement a language toggle (English / Filipino / Bikol) across all pages. Multiple respondents cited this as the most critical improvement for the Tigaon community — particularly for elderly residents and non-English speakers. This directly serves the platform's mission as a community health resource and addresses H2 violations.

02 **Implement Visible Form Submission Feedback**
Add clear success/error toast notifications or confirmation screens upon form submission. Users should immediately know whether their appointment or inquiry was received. This resolves H1 and H9 violations and should improve Q9 ("felt confident") scores.

03 **Enrich the Doctor and Specialist Directory**
Expand each doctor profile to include specialization details, consultation schedules, and real-time availability. Respondents noted the current directory lacked actionable information. This addresses H2 violations and the most commonly cited open-ended feedback issue.

04 **Redesign Homepage with a Clear Visual Hierarchy**
Simplify the homepage by placing the primary call-to-action above the fold and restructuring the service directory into scannable cards. Resolves H8 violations and directly targets the learnability gap in Q10 (avg 2.60).

05 **Add Step-by-Step Onboarding and Help Section**
Introduce a guided walkthrough for first-time users and a persistent FAQ/help section. Respondents suggested a "step-by-step tutorial on how to use the website." Addresses H10 violations and the finding that 12 of 40 respondents still felt they needed technical support.

06 **Enable Back Navigation in Multi-Step Flows**
Refactor the appointment booking flow into a clearly stepped form with progress indicators and a back button at each step. Resolves H3 and H5 violations and reduces the learnability gap for the 7 respondents who found the system harder to navigate than expected.

07 **Expand Platform Coverage and Offline Accessibility**
Design the platform to be usable by people without smartphones through offline-accessible forms or SMS fallback options. This extends the system's reach beyond tech-savvy users and aligns with its public health mission of serving the entire Tigaon community.

Tigaon – Primary Hospital and Clinic Website · Usability Evaluation